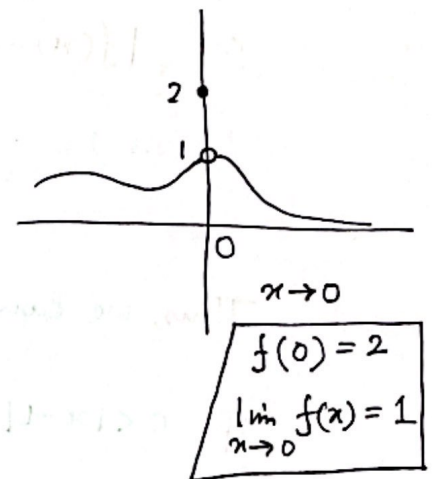


Lecture 1

Definition of Limit:

What is the difference between a defined function at $x=a$ and x is near to a ?

Since $f(a)$ at $x=a$ is a functional value and x approaches to 'a' means $f(x)$ approaches to $f(a)$ which is not the exact value all times (in general).



Definition:

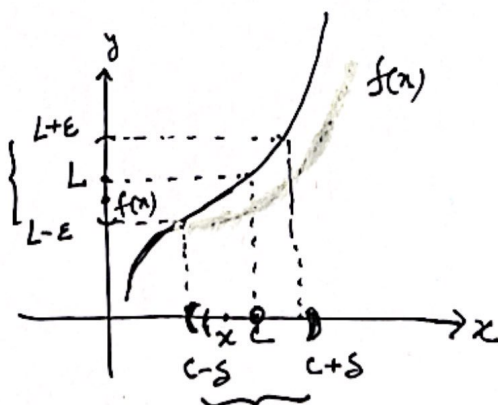
Let $f(x)$ be defined on an open interval about c , except possibly at c itself.

The limit of $f(x)$ as x approaches c is the number L , and write

$$\lim_{x \rightarrow c} f(x) = L.$$

□ if, for every number $\epsilon > 0$, $\exists \delta > 0$ such that

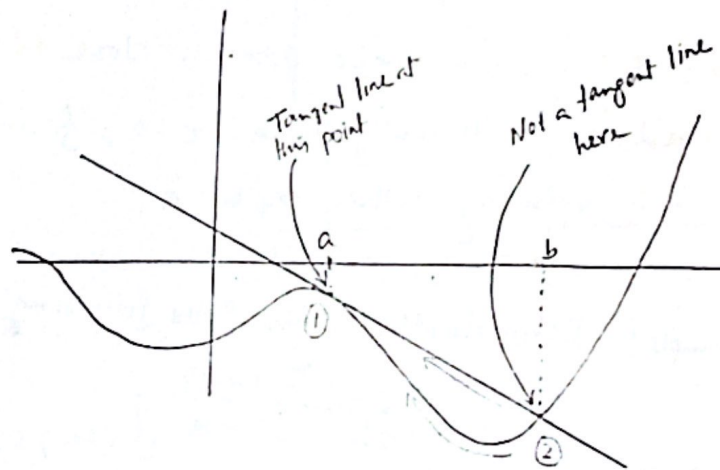
$$|f(x) - L| < \epsilon \text{ whenever } 0 < |x - c| < \delta.$$



Limit:

We will discuss just what a limit tells us about a function as well as how they can be used to get the rate of change of a function as well as the slope of the line tangent to the graph of a function.

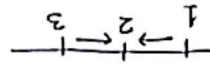
A tangent line to the function $f(x)$ at the point $x=a$ is a line that just touches the graph of the function at the point in the graph, and is 'parallel' (in some way) to the graph at that point.



At the second point shown we will sometimes call the line a secant line when second point approach the first point by using a process and this process (method) we can say the limit of the function of the graph.

We choose values of x that got closer and closer to $x=a$ and we plugged these into the function.

x	$f(x)$	x	$f(x)$
2.5	3.4	1.9	4.157894737
2.1	3.857142857	1.99	4.015075377
2.01	3.985074627	2.999	4.061500750
2.001	3.998500750	2.9999	4.060150008
2.00001	3.999785000	2.99999	4.000015000



Example: Estimate the value of the following limit,

$$\lim_{x \rightarrow 2} \frac{x^2 + 4x - 12}{x^2 - 2x}$$

provided we can make $f(x)$ as close to L as we want for all x sufficiently close to a , from both sides, without actually letting x be a .

$$\lim_{x \rightarrow a} f(x) = L$$

this a

The limit of $f(x)$ is L as x approaches a and with

Definition:

$$\lim_{x \rightarrow a} f(x) = L.$$

some notation for this.

This process is called taking a limit and we have

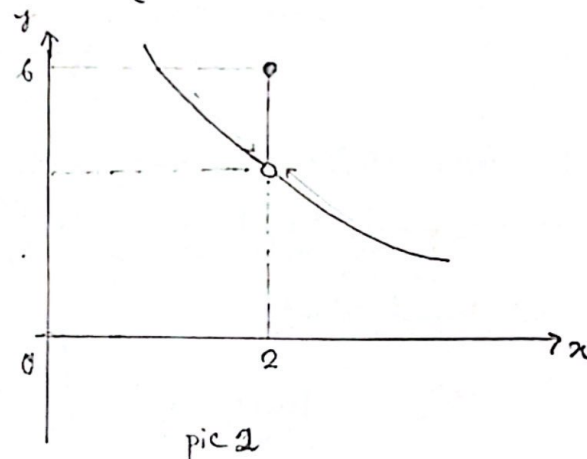
that means the function is going to 4 as x approaches 2, so

$$\lim_{x \rightarrow 2} \frac{x^2 + 4x - 12}{x^2 - 2x} = 4.$$

Example 2: Estimate the value of the following limit,

$\lim_{x \rightarrow 2} g(x)$, where

$$g(x) = \begin{cases} \frac{x^2 + 4x - 12}{x^2 - 2x} & \text{if } x \neq 2 \\ 6 & \text{if } x = 2 \end{cases}.$$

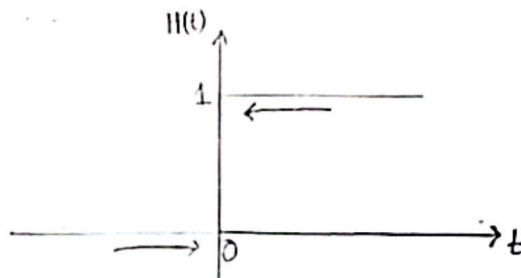


From pic 2 (see) [previous example]

$$\lim_{x \rightarrow 2} g(x) = 4 \quad \text{but when } x = 2, \quad g(2) = 6.$$

Another example called Heaviside or step function.

$$H(t) = \begin{cases} 0 & \text{if } t < 0 \\ 1 & \text{if } t > 0 \end{cases}$$



~~Now we define definition of arbitrary.~~
Now we add Right and Left ^{hand} side of a limit.

Right hand limit

We say $\lim_{x \rightarrow a^+} f(x) = L$

provided we can make $f(x)$ as close to L as we want for all x sufficiently close to ' a ' with $x > a$ without actually letting x be a .

Left hand limit

We say $\lim_{x \rightarrow a^-} f(x) = L$

provided we can make $f(x)$ as close to L as we want for all x sufficiently close to ' a ' with $x < a$ without actually letting x be a .

So, for the previous example,

$$\lim_{t \rightarrow 0^+} H(t) = 1 \quad \text{and} \quad \lim_{t \rightarrow 0^-} H(t) = 0$$

Now we are giving a mathematical definition of a limit.

Definition of a limit:

Given a function $f(x)$ if,

$$\lim_{x \rightarrow a^+} f(x) = \lim_{x \rightarrow a^-} f(x) = L$$

then the normal limit will exist and

$$\lim_{x \rightarrow a} f(x) = L.$$

Properties:

We will assume that $\lim_{x \rightarrow a} f(x)$ and $\lim_{x \rightarrow a} g(x)$ exist and

c is any constant. Then,

$$1. \lim_{x \rightarrow a} [cf(x)] = c \lim_{x \rightarrow a} f(x)$$

$$2. \lim_{x \rightarrow a} [f(x) \pm g(x)] = \lim_{x \rightarrow a} f(x) \pm \lim_{x \rightarrow a} g(x)$$

$$3. \lim_{x \rightarrow a} [f(x)g(x)] = \lim_{x \rightarrow a} f(x) \cdot \lim_{x \rightarrow a} g(x)$$

$$4. \lim_{x \rightarrow a} \left[\frac{f(x)}{g(x)} \right] = \frac{\lim_{x \rightarrow a} f(x)}{\lim_{x \rightarrow a} g(x)}, \quad \text{and} \quad \lim_{x \rightarrow a} g(x) \neq 0.$$

$$5. \lim_{x \rightarrow a} [f(x)]^n = \left[\lim_{x \rightarrow a} f(x) \right]^n \quad \text{e.g.} \quad \lim_{x \rightarrow a} (x^2 + 1)^3 = \left[\lim_{x \rightarrow a} (x^2 + 1) \right]^3$$

Let
Problem: We defined a function

$$f(x) = \begin{cases} \frac{1}{x+2} & , x < -2 \\ x^2 - 5 & , -2 \leq x \leq 3 \\ \sqrt{x+13} & , x > 3 \end{cases}$$

Find the $\lim_{x \rightarrow -2} f(x)$ and $\lim_{x \rightarrow 3} f(x)$ if it exist.

Solution: at $x = -2$:

$$\text{L.H.L.} = \lim_{x \rightarrow -2^-} f(x) = \lim_{x \rightarrow -2^-} \frac{1}{x+2} = -\infty$$

$$\text{R.H.L.} = \lim_{x \rightarrow -2^+} f(x) = \lim_{x \rightarrow -2^+} (x^2 - 5) = -1$$

So, L.H.L. $\neq$ R.H.L. thus $\lim_{x \rightarrow -2} f(x)$ does not exist.

2nd part:

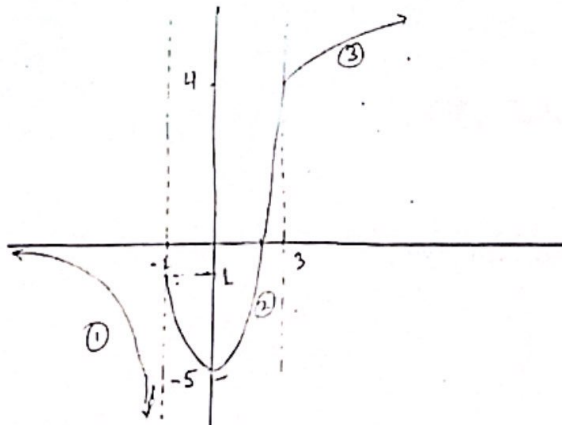
at $x = 3$:

$$\text{L.H.L.} = \lim_{x \rightarrow 3^-} f(x) = \lim_{x \rightarrow 3^-} (x^2 - 5) = 4$$

$$\text{R.H.L.} = \lim_{x \rightarrow 3^+} f(x) = \lim_{x \rightarrow 3^+} \sqrt{x+13} = \sqrt{16} = 4$$

Thus the limit exists at $x = 3$ and $\lim_{x \rightarrow 3} f(x) = 4$

✱



Problem: Estimate the limit: $\lim_{x \rightarrow 0} \frac{x}{|x|}$ if it exists and $x \neq 0$.

Solution: Assume that

$$f(x) = \frac{x}{|x|} = \begin{cases} \frac{x}{x} = 1 & , x > 0 \\ \frac{x}{-x} = -1 & , x < 0 \end{cases}$$

Now, at $x=0$:

$$\text{L.H.L.} = \lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} \frac{x}{-x} = -1$$

$$\text{R.H.L.} = \lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} \frac{x}{x} = 1$$

So, L.H.L. $\neq$ R.H.L. Therefore, $\lim_{x \rightarrow 0} \frac{x}{|x|}$ does not exist. $\times$

Problem: Estimate $\lim_{x \rightarrow 0} x \sin\left(\frac{1}{x}\right)$ if exists.

Solution: Do we use the property of limit such as

$$\lim_{x \rightarrow a} [f(x) \cdot g(x)] = \lim_{x \rightarrow a} f(x) \cdot \lim_{x \rightarrow b} g(x)$$

The answer is 'NO'. Because the properties of limit exist if the individual limit exists. So, we need another way to solve this problem by using a theorem "Squeeze theorem". Next page we write it.

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Example: Show that $\lim_{x \rightarrow 1} (5x-3) = 2$ using ϵ - δ definition.

Solⁿ: Say, $f(x) = 5x-3$, $c = 1$, $L = 2$

By definition, for any given $\epsilon > 0$, we have to find a suitable $\delta > 0$ so that (whenever)
 $0 < |x-1| < \delta$.

$$\text{so, } |f(x) - 2| < \epsilon \Rightarrow |5x-3-2| < \epsilon$$

$$\Rightarrow |5x-5| < \epsilon$$

$$\Rightarrow |x-1| < \frac{\epsilon}{5}$$

Thus, we can take $\delta = \frac{\epsilon}{5}$.

$$\text{If } 0 < |x-1| < \delta = \frac{\epsilon}{5}, \text{ then } |(5x-3)-2| = |5x-5| = 5|x-1| < 5 \cdot \frac{\epsilon}{5} = \epsilon,$$

which proves that $\lim_{x \rightarrow 1} (5x-3) = 2$.

How to Find a δ for a given f , L , c , and $\epsilon > 0$

Steps:

1st step: Solve the inequality $|f(x) - L| < \epsilon$ to find an open interval (a, b) containing c on which the inequality holds for all $x \neq c$.

2nd Step: Find a value of $\delta > 0$ that places the open interval $(c - \delta, c + \delta)$ centered at c inside the interval (a, b) .

The inequality $|f(x) - L| < \epsilon$ will hold for all $x \neq c$ in this δ -interval.

Example: Prove that $\lim_{x \rightarrow 2} f(x) = 4$ if

$$f(x) = \begin{cases} x^2, & x \neq 2 \\ 1, & x = 2 \end{cases}$$

Solⁿ: We want to show that given $\epsilon > 0 \exists \delta > 0$ s.t.

$$|f(x) - 4| < \epsilon \text{ whenever } 0 < |x - 2| < \delta.$$

Solve the inequality $|f(x) - 4| < \epsilon$ to find an open interval containing $x = 2$ on which the inequality holds for all $x \neq 2$.

For $x \neq c = 2$, we have $f(x) = x^2$, we have

$$|x^2 - 4| < \epsilon$$

$$- \epsilon < x^2 - 4 < \epsilon$$

$$4 - \epsilon < x^2 < 4 + \epsilon$$

$$\sqrt{4-\epsilon} < |x| < \sqrt{4+\epsilon}$$

$$\sqrt{4-\epsilon} < x < \sqrt{4+\epsilon}$$

The inequality $|f(x)-4| < \epsilon$ holds for all $x \neq 2$ in the interval

$(\sqrt{4-\epsilon}, \sqrt{4+\epsilon})$, assume $\epsilon < 4$.

2nd step

Now find a value of $\delta > 0$ that places the centered interval $(2-\delta, 2+\delta)$ inside the interval $(\sqrt{4-\epsilon}, \sqrt{4+\epsilon})$

Take δ to be the distance from $x=2$ to the nearer endpoint of $(\sqrt{4-\epsilon}, \sqrt{4+\epsilon})$.

i.e., ~~Take $\delta = \min \{ \sqrt{4-\epsilon}, \sqrt{4+\epsilon} \}$~~

Take $\delta = \min \{ 2 - \sqrt{4-\epsilon}, \sqrt{4+\epsilon} - 2 \}$

$\therefore \forall x$ $|f(x)-4| < \epsilon$ whenever $0 < |x-2| < \delta$, say $\epsilon < 4$.

if $\epsilon \geq 4$ then take $\delta = \min \{ 2, \sqrt{4+\epsilon} - 2 \}$.